Portable 3D Scanner for Digital Reconstruction

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1 Abstract

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2 Introduction

"What?" This paper presents a portable handheld 3D scanner designed for digital reconstruction applications. "So What?" The scanner is made out of off-the-shelf components, including a depth sensor, an Inertial Measurement Unit (IMU), and a microcontroller. A portable 3D scanner is useful for various applications, such as cultural heritage preservation, industrial inspection, and virtual reality content creation.

3 State of the Art

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8 Brainstorming

Ficheiro de armazenamento da point cloud Transmissão da point cloud para o computador Interface de visualização da point cloud Programa para reconstrução da point cloud em malha Através de um ponto no espaço, determinar a posição do scanner (Geometria que permite a colocação do ponto no espaço) Montagem de um protótipo handheld scanner (Esqueleto apenas com buracos para os sensores e a placa)

References

[1] Dingkang Wang, Connor Watkins, and Huikai Xie. "MEMS Mirrors for LiDAR: A Review". In: *Micromachines* 11.5 (May 2020), p. 456. ISSN: 2072-666X. DOI: 10.3390/mi11050456. URL: https://www.mdpi.com/2072-666X/11/5/456 (visited on 10/08/2026).